

ASHTONE ONYANGO

Mid-Senior AI Engineer

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PROFESSIONAL SUMMARY

Mid-Senior AI Engineer with 6+ years designing, training, and deploying production AI systems, from transformer models trained from scratch to multi-agent orchestration platforms serving enterprise clients at scale. Hands-on across the full ML stack: LLM fine-tuning (RLHF/SFT), RAG pipeline architecture, conversational AI design, and cloud-native MLOps on GCP and AWS. Proven track record moving AI prototypes to production, with deep experience in prompt engineering, agentic workflow automation, and building evaluation frameworks that drive continuous improvement. Comfortable in client-facing technical delivery and cross-functional collaboration.

SKILLS & COMPETENCIES

- **LLM & GenAI:** Gemini (Pro/Ultra), GPT-4, Claude, Vertex AI, Dialogflow CX, CCAI/CXAS, Agent Assist, Generative Knowledge Assist, Prompt Engineering, RAG, RLHF, SFT
 - **Agentic Frameworks:** LangChain, CrewAI, LangGraph, CXAS Playbooks, multi-agent orchestration, tool-use, grounding, reasoning pipelines
 - **Cloud & Infrastructure:** Google Cloud Platform (Vertex AI, Cloud Run, BigQuery, Pub/Sub), AWS (SageMaker, Lambda, S3), Terraform, Kubernetes, Docker, CI/CD (GitHub Actions, Jenkins)
 - **Telephony & CX Integration:** Dialogflow CX, CCAI Insights, CCaaS platforms (Genesys, Cisco/Avaya architecture), SIP/WebRTC familiarity, Salesforce & Zendesk API integration
 - **LLM Internals:** Attention mechanisms (MHA, FlashAttention), KV caches, tokenizers, inference optimization, distributed training (DDP), LoRA/QLoRA fine-tuning
 - **Data & Vector Systems:** Pinecone, FAISS, Vertex AI Search, BigQuery, PostgreSQL, Redis, Apache Spark, Kafka, Airbyte
 - **Evaluation & Agent Ops:** LLM evaluation design, automated scoring pipelines, latency/accuracy/safety benchmarking, CCAI Insights, ground-truth verification, reward function design
 - **Languages:** Python (advanced), JavaScript/Node.js, SQL, Bash
 - **Certifications:** AWS ML Engineer (Udacity), IBM AI Analyst Mastery, Microsoft IoT/AI, Data Engineer (DataCamp), Full-Stack Developer (Udacity)
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WORK EXPERIENCE

iMerit Technology | RL Environment & LLM Evaluation Engineer (Contract) Nov 2025 – Mar 2026 | Remote, US

- Designed and built high-precision RL environments and annotation workflows for GenAI training pipelines targeting GPT-4, Claude, and Gemini, including boundary testing, failure mode identification, and reward-hacking analysis.
- Engineered computationally intensive STEM and data science problem sets simulating real-world scientific workflows, used as ground-truth RL training tasks for LLM fine-tuning.

- Developed non-trivial reasoning chains and adversarial evaluation cases; implemented automated judge logic equivalent to continuous RL reward signals using Python (NumPy, Pandas) and SQL scoring pipelines.
- Deployed and managed reproducible Docker environments and CI/CD pipelines on AWS and GCP for the delivery of high-fidelity training datasets at scale.
- Served as the technical liaison between expert annotators and AI systems teams, improving annotation strategy and data quality pipelines for RLHF and SFT workflows.

Wan AI Labs | Lead AI Engineer (Agentic Systems & LLM Deployment) Oct 2024 – Sep 2025 |
Nairobi, Kenya

- Led the AI engineering division, architecting and deploying 20+ production-grade agentic systems using GPT-4, Claude, and Gemini across healthcare, legal, and enterprise operations — driving 40% productivity gains and 60% reduction in manual operations.
- Built and deployed multi-agent orchestration platforms using LangChain and CrewAI; designed reasoning pipelines with tool-use, grounding, and retrieval-augmented generation (RAG) over domain-specific knowledge bases achieving 95% information accuracy.
- Designed and maintained RAG pipelines over structured and unstructured enterprise knowledge bases using Pinecone and FAISS, enabling real-time grounded responses in client-facing AI agents deployed via REST and GraphQL APIs.
- Implemented prompt engineering and orchestration layers for brand voice consistency, safety guardrails, and accuracy optimization across multi-turn conversational workflows integrated with Salesforce and Zendesk CRM systems.
- Established Agent Ops practices and evaluation frameworks measuring agent latency, accuracy, hallucination rate, and customer satisfaction, iterating models based on CCAI-style insights and structured human feedback, directly applicable to RLHF pipeline design.
- Managed full deployment lifecycle on GCP and AWS: model selection, containerization (Docker/Kubernetes), CI/CD automation (GitHub Actions), and production monitoring with reproducible infrastructure-as-code environments.

Rightsify Group LLC | AI Engineer — Audio Generative LLMs Jun 2023 – Aug 2024 | Pasadena, CA (Remote)

- Architected and trained custom transformer LLMs from scratch for music generation, serving thousands of daily users, covering full training pipelines: data preprocessing, model architecture design, distributed training on AWS with PyTorch and HuggingFace.
- Implemented reinforcement learning feedback loops and reward functions during model training, achieving a 20% improvement in output quality, directly analogous to RLHF reward model design and continuous improvement cycles.
- Designed distributed training pipelines using PyTorch DDP on AWS SageMaker across multi-GPU instances, reducing training time by 60% while processing 1M+ audio samples through custom preprocessing workflows.
- Built continuous model evaluation and performance monitoring pipelines, automated scoring of model outputs against ground-truth benchmarks, equivalent to LLM eval frameworks measuring accuracy, safety, and latency.
- Implemented CI/CD for ML model deployment using Docker and Kubernetes, reducing deployment time from 4 hours to 15 minutes in fully containerized, reproducible environments with Terraform-managed infrastructure.

- Automated 10TB+ dataset migration pipelines (AWS S3 → GCP), establishing scalable MLOps infrastructure for model serving across Paperspace and DigitalOcean.

Zuri Health | Software Developer *Jan 2023 – Apr 2023 | Nairobi, Kenya*

- Built a conversational AI triage system using NLP and RAG architecture on GCP, serving 5,000+ users with 92% diagnostic accuracy and a 30% reduction in query resolution time, a production Dialogflow-style system integrated with healthcare knowledge bases.
- Designed a semantic search infrastructure over medical information using vector embeddings, improving answer relevance by 40% through optimized retrieval pipelines.
- Collaborated with platform engineers to design analytics pipelines extracting actionable insights from multi-country user interaction data.

Google Crowdsourcing | AI Facilitator / Trainer (Part-time)

Aug 2021 – Aug 2025 |

Nairobi, Kenya

- Facilitated 11+ hands-on ML workshops covering LLM fine-tuning, NLP pipeline design, and cloud ML deployment (AWS SageMaker, GCP Vertex AI) for 33+ university students across Africa.
- Managed a community of 5,000+ AI contributors, training participants on deploying production ML models on GCP and AWS cloud infrastructure.

Analytics Vidhya | Technical Writer — Data (Seasonal)

Sep 2022 – Dec 2022 | Gurgaon,

India (Remote)

- Published 3 tutorial articles on ETL pipeline design over OLAP/OLTP databases; implemented Flask application backends integrating PostgreSQL APIs hosted locally and on cloud infrastructure.

Upwork | Machine Learning Engineer (Freelance)

Feb 2021 – Sep 2022 | Nairobi,

Kenya

- Delivered 33+ custom AI solutions leveraging AWS and Azure ML services, semantic search, NLP systems, and LLM-powered data processing pipelines for diverse clients.
- Built production-ready GraphQL and RESTful APIs with Express and React frontends, integrating LLM capabilities for intelligent data processing and automation.
- Developed distributed data processing pipelines using Apache Spark, improving data ingestion speed by 65%.

Teens in AI | Machine Learning Trainer

Aug 2019 – Sep 2019 | London, UK

- Coached teams on Descriptive and Predictive Analytics using Azure Machine Learning; guided ~10 students in applying ML to analyze affordable healthcare access across Sub-Saharan Africa.

EDUCATION

Kenyatta University, Nairobi, Kenya | BSc Biomedical Engineering

Relevant: Health Information Systems, Bioinformatics, Programming, Electronics, Database Systems.

Certifications:

- AWS Machine Learning Engineer Nanodegree; Udacity.
- Data Engineer Track; DataCamp.
- Full-Stack Developer Nanodegree; Udacity.
- Information Security Certification; FreeCodeCamp.
- IBM AI Analyst Mastery Award & Explorer Award (2019).
- Microsoft IoT & AI Certifications.

KEY PROJECTS

TAF: AI Caregiver Support Ecosystem

- Custom RAG architecture over medical and legal knowledge bases, serving 1,000+ caregivers across Sub-Saharan Africa with 95% information accuracy. Production LLM fine-tuning, retrieval pipeline design, and evaluation framework — directly analogous to enterprise CXAS deployments.

DriveSmart AI

- WhatsApp-Powered AI Assistant that gives every driving school learner a 24/7 intelligent assistant trained on the school's own materials. Learners ask via text, voice, or photo and get accurate answers instantly, solving the 80% knowledge retention problem after learners leave the classroom.

CDIE Smart Lab: LLM-Powered SOP Automation

- LLM document processing system automating understanding and assessment of Lab SOPs for 10,000+ university users, achieving 75% faster lab preparation. Judge-style correctness evaluation pipeline at its core.

Kujia Jobs: Semantic Job Matching Engine

- Custom NLP semantic search engine matching 5,000+ job seekers to opportunities with 90% accuracy, handling 1,000+ daily queries on AWS Lambda with collaborative filtering.

Pneumonia Detection & Monitoring in Infants

- Neural network trained from scratch on 5,000+ IoT sensor data points, achieving 80% diagnostic accuracy. Awarded at Huawei Global Innovator Summit, Johannesburg, 2022.